

Assignment Number 2.2 LastName: Dash First: Altankhuyag Initial DA.pdf

Source: 6535 Ivy Hill Dr, Mclean VA

Link for project photo access: <https://www.redfin.com/VA/McLean/6535-Ivy-Hill-Dr-22101/home/9474828>

photo: Framing Photo



Materials:

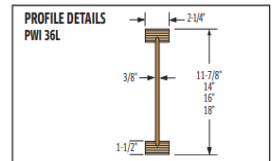
1. Floor Joist : I-joist - Wood
2. LVL (glue-laminated) Beam
3. Metal wind bracing
4. OSB floor panel
5. 2x4 wall studs & Jack, POSTS,
6. Portal Framing LVL header

1. Definition: I -floor joist PWI 36 (technical specs) - used 16" flange 2 1/4" **Floor joists** are the horizontal structural members that support the flooring system of a building. They span between the foundation walls or beams, providing the framework for the subfloor

and finished flooring.

DESIGN VALUES

Series	Depth	Weight (plf)	Moment (lb-ft)	EI (x 10 ⁶) (lb-in ²)	K (x 10 ⁶) (lb-ft/in)	Shear (lbs)
PWI 36L	11-7/8"	3.1	6445	429	0.468	1615
	14"	3.4	7755	622	0.55	1830
	16"	3.6	8995	836	0.625	2020
	18"	3.9	10135	1082	0.7	2185

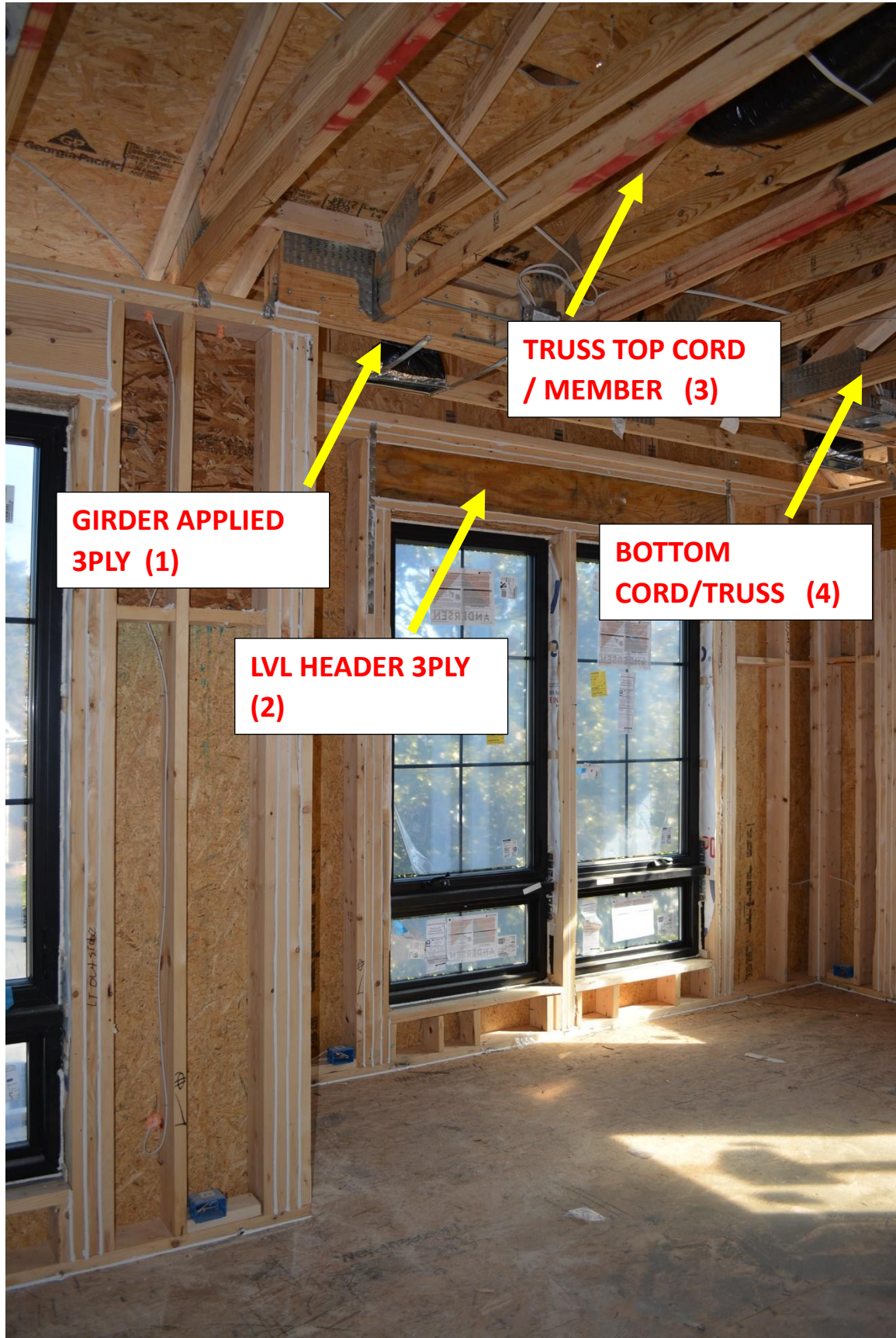


- Definition: LVL (Laminated Veneer Lumber) glue laminated beams are engineered wood products that offer exceptional strength, stability, and versatility. They are composed of multiple thin layers of wood veneers that are bonded together with a strong adhesive. This process creates a beam that is significantly stronger than traditional solid wood beams.
Structural components: Beams, headers, joists
- Metal wind bracing is a structural element used to provide lateral stability to buildings and other structures, particularly in areas prone to high winds. It consists of metal members, such as angles, channels, or wide-flange beams, that are connected to the building's frame to resist forces caused by wind.
- Oriented Strand Board (OSB) panels are a popular choice for flooring applications due to their strength, affordability, and versatility. They are engineered wood products made from wood strands that are bonded together with a resin adhesive.
- Portal frame LVL beams** are engineered wood structures that combine the strength and stability of LVL (Laminated Veneer Lumber) with the efficient design of portal frames. This combination makes them a popular choice for various construction applications, including residential, commercial, and industrial buildings.





- **Partition Wall:** A non-load-bearing wall that separates spaces within a building.
 - **YELLOW** – PARTITION WALL & STUDS

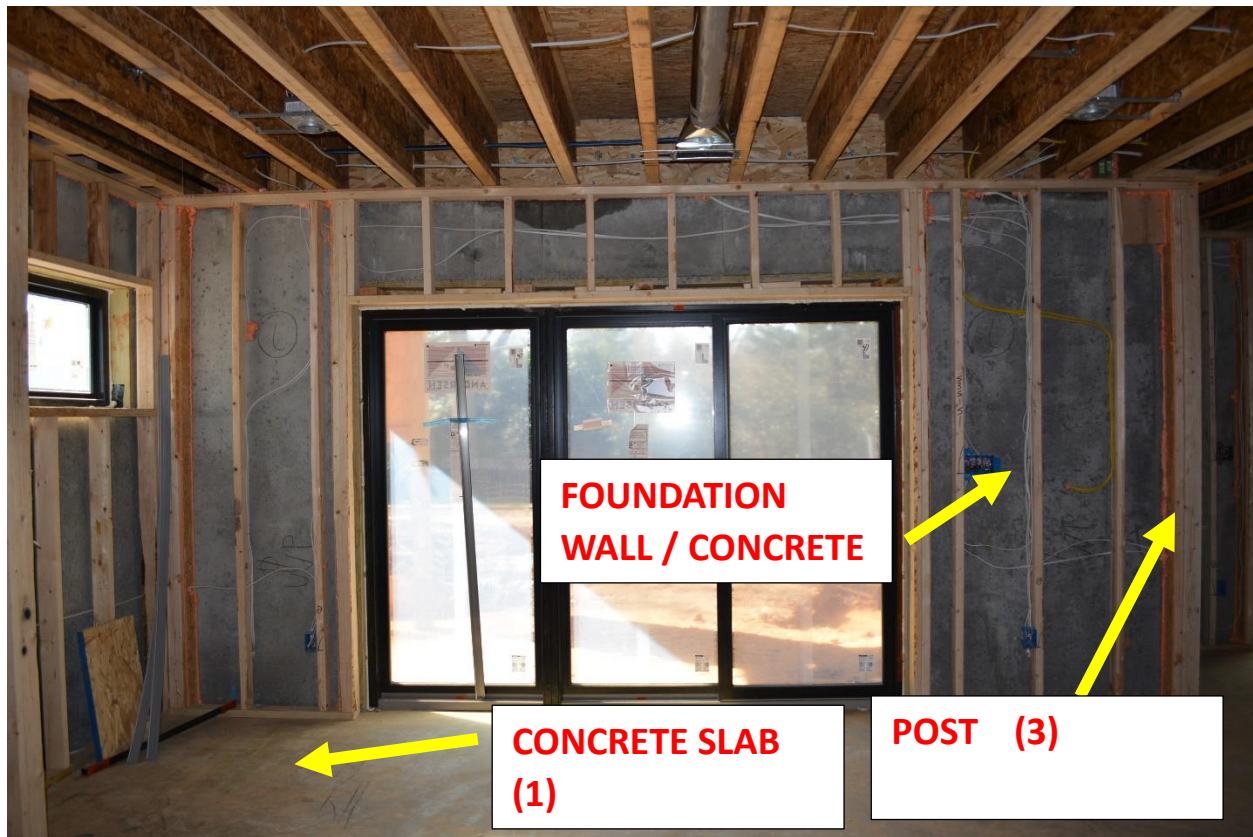


**TRUSS TOP CORD
/ MEMBER (3)**

**GIRDER APPLIED
3PLY (1)**

**BOTTOM
CORD/TRUSS (4)**

**LVL HEADER 3PLY
(2)**



Columns and Posts

- **Column:** A vertical structural member that supports loads from above.
- **Stud:** A small, vertical member used in walls for framing.

Foundations

- **Footing:** A concrete base that distributes the load of a structure to the soil.
- **Pile Foundation:** A foundation that uses piles driven into the ground to support a structure.
- **Mat Foundation:** A foundation that covers a large area to support a heavy structure.

Roofing Components

- **Girder:** A large beam that supports other beams or joists.
- **Cantilever:** A beam that extends beyond its supports.
- **Rafter:** A sloping member that supports the roof covering.
- **Ridge Beam:** The horizontal beam at the top of a roof.
- **Header Beam:** A beam that supports the ends of joists or rafters.

Walls and Bracing

- **Partition Wall:** A non-load-bearing wall that separates spaces within a building.
- **Brace:** A diagonal member that provides stability to a structure.
- **Base Plate:** A metal plate that connects a column to its foundation.
- **Retaining Wall:** A wall that supports soil at a different level.
- **Metal Deck with Concrete Fill:** A composite floor system consisting of a metal deck with concrete poured on top.
- **Truss:** A framework of interconnected members that provides structural support.

